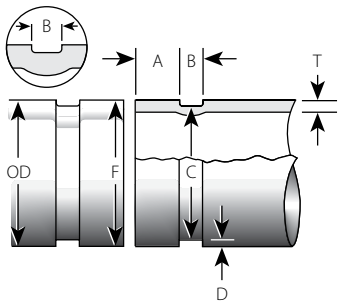
Victaulic® Original Groove System (OGS) Specifications

1.0 DIMENSIONS

OGS ROLL GROOVE SPECIFICATIONS FOR CARBON STEEL PIPE AND ALL MATERIALS GROOVED WITH STANDARD AND RX ROLLS



Exaggerated for Clarity

Nominal Size inches DN	inches/millimeters													
	Pipe Outside Diameter[1]			Gasket Seat "A"[2]			Groove Width "B"[3]			Groove Diameter "C"[4]		Groove Depth "D"[5] (ref.)	Min. Nom. Wall Thick. "T"[6]	Max. Allow. Flare Dia. "F"[7]
	Actual	Max.	Min.	Basic	Max.	Min.	Basic	Max.	Min.	Max.	Min.			
¾ DN20	1.050 26.9	1.060 26.9	1.040 26.4	0.625 15.9	0.656 16.7	0.594 15.1	0.281 7.1	0.312 7.9	0.250 6.4	0.938 23.8	0.923 23.4	0.056 1.5	0.049 1.2	1.15 29.2
1 DN25	1.315 33.7	1.328 33.7	1.302 33.1	0.625 15.9	0.656 16.7	0.594 15.1	0.281 7.1	0.312 7.9	0.250 6.4	1.190 30.2	1.175 29.9	0.063 1.6	0.049 1.2	1.43 36.3
1 ¼ DN32	1.660 42.4	1.676 42.6	1.644 41.8	0.625 15.9	0.656 16.7	0.594 15.1	0.281 7.1	0.312 7.9	0.250 6.4	1.535 39.0	1.520 38.6	0.063 1.6	0.049 1.2	1.77 45.0
1 ½ DN40	1.900 48.3	1.919 48.7	1.881 47.8	0.625 15.9	0.656 16.7	0.594 15.1	0.281 7.1	0.312 7.9	0.250 6.4	1.775 45.1	1.760 44.7	0.063 1.6	0.049 1.2	2.01 51.1
	2.244 57.0	2.267 57.6	2.222 56.4	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	2.118 53.8	2.102 53.4	0.063 1.6	0.049 1.2	2.35 59.7
2 DN50	2.375 60.3	2.399 60.9	2.351 59.7	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	2.250 57.2	2.235 56.8	0.063 1.6	0.049 1.2	2.48 63.0
2 ½	2.875 73.0	2.904 73.8	2.846 72.3	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	2.720 69.1	2.702 68.6	0.078 2.0	0.078 2.0	2.98 75.7
DN65	3.000 76.1	3.030 77.0	2.970 75.4	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	2.845 72.3	2.827 71.8	0.078 2.0	0.078 2.0	3.10 78.7
3 DN80	3.500 88.9	3.535 89.8	3.469 88.1	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	3.344 84.9	3.326 84.5	0.078 2.0	0.078 2.0	3.60 91.4
3 ½ DN90	4.000 101.6	4.040 102.6	3.969 100.8	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	3.834 97.4	3.814 96.9	0.083 2.1	0.078 2.0	4.10 104.1
	4.250 108.0	4.293 109.0	4.219 107.2	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	4.084 103.7	4.064 103.2	0.083 2.1	0.078 2.0	4.35 110.5
4 DN100	4.500 114.3	4.545 115.4	4.469 113.5	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	4.334 110.1	4.314 109.6	0.083 2.1	0.078 2.0	4.60 116.8
4 ½	5.000 127.0	5.050 128.3	4.969 126.2	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	4.834 122.8	4.814 122.3	0.083 2.1	0.078 2.0	5.10 129.5
	5.250 133.0	5.303 134.7	5.219 132.6	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	5.084 129.1	5.064 128.6	0.083 2.1	0.078 2.0	5.35 135.9
DN125	5.500 139.7	5.556 141.1	5.469 138.9	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	5.334 135.5	5.314 135.0	0.083 2.1	0.078 2.0	5.60 142.2
5	5.563 141.3	5.619 142.7	5.532 140.5	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	5.395 137.0	5.373 136.5	0.084 2.1	0.078 2.0	5.66 143.8
	6.000 152.4	6.056 153.8	5.969 151.6	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	5.830 148.1	5.808 147.5	0.085 2.2	0.078 2.0	6.10 154.9
	6.250 159.0	6.313 160.4	6.219 158.0	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	6.032 153.2	6.002 152.5	0.109 2.8	0.109 2.8	6.35 161.3
	6.500 165.1	6.563 166.7	6.469 164.3	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	6.330 160.8	6.308 160.2	0.085 2.2	0.078 2.0	6.60 167.6

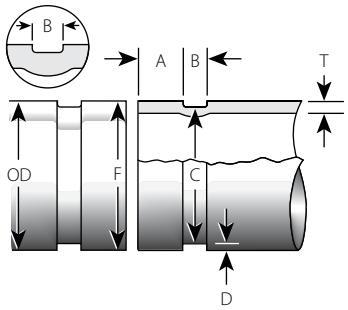
REFERENCE NOTES ON PAGE 3

ALWAYS REFER TO ANY NOTIFICATIONS AT THE END OF THIS DOCUMENT REGARDING PRODUCT INSTALLATION, MAINTENANCE OR SUPPORT.

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1.0 DIMENSIONS (CONTINUED)

OGS ROLL GROOVE SPECIFICATIONS FOR CARBON STEEL PIPE AND ALL MATERIALS GROOVED WITH STANDARD AND RX ROLLS



Exaggerated for Clarity

Nominal Size inches DN	inches/millimeters													
	Pipe Outside Diameter[1]			Gasket Seat "A"[2]			Groove Width "B"[3]			Groove Diameter "C"[4]		Groove Depth "D"[5] (ref.)	Min. Nom. Wall Thick. "T"[6]	Max. Allow. Flare Dia. "F"[7]
	Actual	Max.	Min.	Basic	Max.	Min.	Basic	Max.	Min.	Max.	Min.			
6 DN150	6.625 168.3	6.688 169.9	6.594 167.5	0.625 15.9	0.656 16.7	0.594 15.1	0.344 8.7	0.375 9.5	0.313 8.0	6.455 164.0	6.433 163.4	0.085 2.2	0.078 2.0	6.73 170.9
	8.000 203.2	8.063 204.8	7.969 202.4	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	7.816 198.5	7.791 197.9	0.092 2.4	0.109 2.8	8.17 207.5
# 8 DN200	8.515 216.3	8.578 217.9	8.484 215.5	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	8.331 211.6	8.306 211.0	0.092 2.4	0.109 2.8	8.69 220.7
	8.625 219.1	8.688 220.7	8.594 218.3	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	8.441 214.4	8.416 213.8	0.092 2.4	0.109 2.8	8.80 223.5
	10.000 254.0	10.063 255.6	9.969 253.2	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	9.812 249.2	9.785 248.5	0.094 2.4	0.134 3.4	10.17 258.3
# 10 DN250	10.528 267.4	10.591 269.0	10.497 266.6	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	10.340 262.6	10.313 262.0	0.094 2.4	0.134 3.4	10.70 271.8
	10.750 273.0	10.813 274.7	10.719 272.3	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	10.562 268.3	10.535 267.6	0.094 2.4	0.134 3.4	10.92 277.4
	12.000 304.8	12.063 306.4	11.969 304.0	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	11.781 299.2	11.751 298.5	0.109 2.8	0.156 4.0	12.17 309.1
# 12 DN300	12.539 318.5	12.602 320.1	12.508 317.7	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	12.321 313.0	12.291 312.2	0.109 2.8	0.156 4.0	12.71 322.8
	12.750 323.9	12.813 325.5	12.719 323.1	0.750 19.1	0.781 19.8	0.719 18.3	0.469 11.9	0.500 12.7	0.438 11.1	12.531 318.3	12.501 317.5	0.109 2.8	0.156 4.0	12.92 328.2
14* DN350	14.000 355.6	14.063 357.2	13.969 354.8	0.938 23.8	0.969 24.6	0.907 23.0	0.469 11.9	0.500 12.7	0.438 11.1	13.781 350.0	13.751 349.3	0.109 2.8	0.156 4.0	14.16 359.7
	14.843 377.0	14.937 379.4	14.811 376.2	0.938 23.8	0.969 24.6	0.907 23.0	0.469 11.9	0.500 12.7	0.438 11.1	14.611 371.1	14.581 370.4	0.116 2.9	0.177 4.5	15.00 381.0
15 DN375	15.000 381.0	15.063 382.6	14.969 380.2	0.938 23.8	0.969 24.6	0.907 23.0	0.469 11.9	0.500 12.7	0.438 11.1	14.781 375.4	14.751 374.7	0.109 2.8	0.165 4.2	15.16 385.1
16* DN400	16.000 406.4	16.063 408.0	15.969 405.6	0.938 23.8	0.969 24.6	0.907 23.0	0.469 11.9	0.500 12.7	0.438 11.1	15.781 400.8	15.751 400.1	0.109 2.8	0.165 4.2	16.16 410.5
	16.772 426.0	16.866 428.4	16.740 425.2	0.938 23.8	0.969 24.6	0.907 23.0	0.469 11.9	0.500 12.7	0.438 11.1	16.514 419.5	16.479 418.6	0.129 3.3	0.177 4.5	16.93 430.0
18* DN450	18.000 457.2	18.063 458.8	17.969 456.4	1.000 25.4	1.031 26.2	0.969 24.6	0.469 11.9	0.500 12.7	0.438 11.1	17.781 451.6	17.751 450.9	0.109 2.8	0.188 4.8	18.16 461.3
	18.898 480.0	18.992 482.4	18.867 479.2	1.000 25.4	1.031 26.2	0.969 24.6	0.469 11.9	0.500 12.7	0.438 11.1	18.626 473.1	18.591 472.2	0.136 3.5	0.236 6.0	19.06 484.1
20* DN500	20.000 508.0	20.063 509.6	19.969 507.2	1.000 25.4	1.031 26.2	0.969 24.6	0.469 11.9	0.500 12.7	0.438 11.1	19.781 502.4	19.751 501.7	0.109 2.8	0.188 4.8	20.16 512.1
	20.866 530.0	20.960 532.4	20.835 529.2	1.000 25.4	1.031 26.2	0.969 24.6	0.469 11.9	0.500 12.7	0.438 11.1	20.572 522.5	20.537 521.6	0.147 3.7	0.236 6.0	21.03 534.2
22* DN550	22.000 558.8	22.063 560.4	21.969 558.0	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	21.656 550.1	21.626 549.3	0.172 4.4	0.188 4.8	22.20 563.9
	22.835 580.0	22.929 582.4	22.803 579.2	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	22.488 571.2	22.457 570.4	0.172 4.4	0.276 7.0	23.03 585.0

REFERENCE NOTES ON PAGE 3

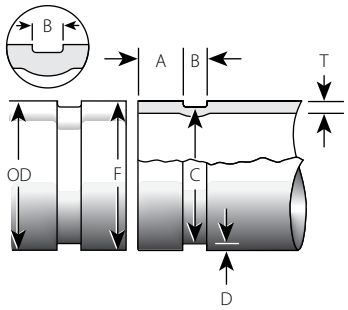
Applies to JIS metric pipe sizes 200A, 250A, and 300A, respectively (JIS Specification G 3452; G 3454).

* OGS roll groove specifications. For Advanced Groove System (AGS) roll groove specifications in these sizes, refer to Victaulic publication 25.09, which can be downloaded at victaulic.com.

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1.0 DIMENSIONS (CONTINUED)

OGS ROLL GROOVE SPECIFICATIONS FOR CARBON STEEL PIPE AND ALL MATERIALS GROOVED WITH STANDARD AND RX ROLLS



Exaggerated for Clarity

Nominal Size inches DN	inches/millimeters													
	Pipe Outside Diameter[1]			Gasket Seat "A"[2]			Groove Width "B"[3]			Groove Diameter "C"[4]		Groove Depth "D"[5] (ref.)	Min. Nom. Wall Thick. "T"[6]	Max. Allow. Flare Dia. "F"[7]
	Actual	Max.	Min.	Basic	Max.	Min.	Basic	Max.	Min.	Max.	Min.			
24*	24.000	24.063	23.969	1.000	1.031	0.969	0.500	0.531	0.469	23.656	23.626	0.172	0.218	24.20
DN600	609.6	611.2	608.8	25.4	26.2	24.6	12.7	13.5	11.9	600.9	600.1	4.4	5.5	614.7
	24.803	24.897	24.772	1.000	1.031	0.969	0.500	0.531	0.469	24.459	24.424	0.172	0.276	25.00
	630.0	632.4	629.2	25.4	26.2	24.6	12.7	13.5	11.9	621.3	620.4	4.4	7.0	635.0

* OGS roll groove specifications. For Advanced Groove System (AGS) roll groove specifications in these sizes, refer to Victaulic publication 25.09, which can be downloaded at victaulic.com.

NOTES FOR OGS ROLL GROOVE SPECIFICATIONS FOR CARBON STEEL PIPE AND ALL MATERIALS GROOVED WITH STANDARD AND RX ROLLS

The roll grooving process does not remove any metal. The groove is cold formed in the pipe through use of a Victaulic roll grooving tool that is equipped with a properly matched roll set for the pipe material and size.

The Victaulic roll groove contains rounded corners that reduce the available pipe-end movement (expansion, contraction, and deflection). For couplings installed on roll grooved pipe, reference the applicable product publication for allowable pipe end separation and deflection values.

The coating thickness applied to the gasket sealing surface and within the groove on the roll-grooved pipe exterior shall not exceed 0.010 inch/0.25 mm. This pipe coating thickness will affect the roll groove specifications listed in the tables on pages 1 – 3. Allowances shall be made for the following:

- Pipe Outside Diameter, Gasket Seat "A", Groove Diameter "C", and Maximum Allowable Flare Diameter "F" will be INCREASED by 0.020 inch/0.50 mm.
- Groove Width "B" will be REDUCED by 0.020 inch/0.50 mm.

In addition, coatings that are applied to the interior surfaces of Victaulic grooved- and plain-end pipe couplings shall not exceed 0.010 inch/0.25 mm. This includes the bolt pad mating surfaces.

1 Outside diameter: Nominal Pipe Size (ANSI B36.10) and Basic Metric Pipe Size (ISO 4200) – The average pipe outside diameter shall not vary from the specifications listed in the table above. Maximum allowable pipe ovality shall not vary by more than 1%. Greater variations between the major and minor diameters will result in difficult coupling assembly.
Any internal and external weld beads or seams shall be ground flush to the pipe surface. The inside diameter of the pipe end shall be cleaned to remove coarse scale, dirt, and other foreign material that might interfere with or damage grooving rolls. The front edge of the pipe end shall be uniform with no concave/convex surface features that will cause improper grooving roll tracking and result in difficulties during coupling assembly. The outside diameter of roll grooved pipe shall not vary more than the tolerance listed.
Victaulic recommends square-cut pipe. Beveled-end pipe may be used, provided that the wall thickness is standard wall (ANSI B36.10) or less and that the bevel meets ANSI B16.25 (37 1/2°) or ASTM A-53 (30°). NOTE: Roll grooving beveled-end pipe may result in unacceptable flare. For OGS, the maximum allowable tolerance from square-cut pipe ends (measured from the true square line) is: 1/32"/0.8 mm for 3/4 – 3 1/2"/DN20 – DN90 sizes and 1/16"/1.6 mm for 4 – 24"/DN100 – DN600 sizes. **Square-cut pipe SHALL be used with Victaulic products containing Flush-Seal™ gaskets.**

2 "A" Dimension: The "A" dimension, or the distance from the pipe end to the groove, identifies the gasket seating area. This area between the groove and the pipe end shall be generally free from indentations, projections, weld seam anomalies, and roll marks to ensure a leak-tight seal. All oil, grease, loose paint, rust, scale, dirt, and cutting particles shall be removed.

3 "B" Dimension: The "B" dimension, or groove width, controls expansion, contraction, and angular deflection of flexible couplings by the distance it is located from the pipe and its width in relation to the coupling housings' "key" width. The bottom of the groove shall be free from loose paint, rust, scale, dirt, and cutting particles that may interfere with proper coupling assembly.

4 "C" Dimension: The "C" dimension is the average diameter at the base of the groove. This dimension shall be within the diameter's tolerance and concentric with the OD for proper coupling fit. The groove shall be of uniform depth for the entire pipe circumference.

5 "D" Dimension: The "D" dimension is the normal depth of the groove and is a reference for a "trial groove" only. Variations in pipe OD affect this dimension and shall be altered, if necessary, to keep the "C" dimension within tolerance. The groove diameter shall conform to the "C" dimension described above.

6 "T" Dimension: The "T" dimension is the lightest grade (minimum nominal wall thickness) of pipe that is suitable for roll grooving. Pipe that is less than the minimum nominal wall thickness may be adapted for Victaulic couplings by using *Vic-Ring* Adapters. *Vic-Ring* Adapters can be used in the following situations (contact Victaulic for details):

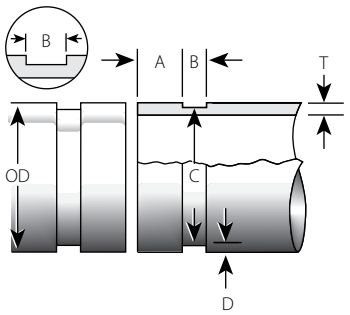
- When pipe is less than the minimum nominal wall thickness suitable for roll grooving
- When pipe outside diameter is too large to roll or cut groove
- When pipe is used in abrasive services

7 "F" Dimension (Roll Groove Only): Maximum allowable pipe-end flare diameter is measured at the extreme pipe-end diameter. **NOTE:** This applies to average (pi tape) and single-point readings.

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1.1 DIMENSIONS

OGS CUT GROOVE SPECIFICATIONS FOR CARBON STEEL AND OTHER IPS PIPE



Exaggerated for Clarity

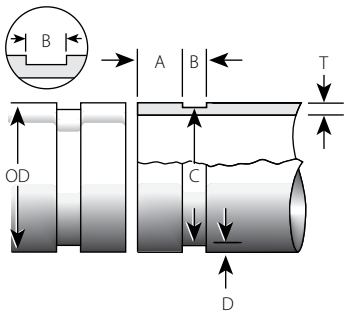
Nominal Size inches DN	inches/millimeters												
	Pipe Outside Diameter[1]			Gasket Seat "A"[2]			Groove Width "B"[3]			Groove Diameter "C"[4]		Groove Depth "D"[5] (ref.)	Min. Nom. Wall Thick. "T"[6]
	Actual	Max.	Min.	Basic	Max.	Min.	Basic	Max.	Min.	Max.	Min.		
¾	1.050	1.060	1.040	0.625	0.656	0.594	0.313	0.344	0.282	0.938	0.923	0.056	0.113
DN20	26.9	26.9	26.4	15.9	16.7	15.1	8.0	8.7	7.2	23.8	23.4	1.5	2.9
1	1.315	1.328	1.302	0.625	0.656	0.594	0.313	0.344	0.282	1.190	1.175	0.063	0.133
DN25	33.7	33.7	33.1	15.9	16.7	15.1	8.0	8.7	7.2	30.2	29.9	1.6	3.4
1 ¼	1.660	1.676	1.644	0.625	0.656	0.594	0.313	0.344	0.282	1.535	1.520	0.063	0.140
DN32	42.4	42.6	41.8	15.9	16.7	15.1	8.0	8.7	7.2	39.0	38.6	1.6	3.6
1 ½	1.900	1.919	1.881	0.625	0.656	0.594	0.313	0.344	0.282	1.775	1.760	0.063	0.145
DN40	48.3	48.7	47.8	15.9	16.7	15.1	8.0	8.7	7.2	45.1	44.7	1.6	3.7
	2.244	2.267	2.222	0.625	0.656	0.594	0.313	0.344	0.282	2.118	2.102	0.063	0.157
	57.0	57.6	56.4	15.9	16.7	15.1	8.0	8.7	7.2	53.8	53.4	1.6	4.0
2	2.375	2.399	2.351	0.625	0.656	0.594	0.313	0.344	0.282	2.250	2.235	0.063	0.154
DN50	60.3	60.9	59.7	15.9	16.7	15.1	8.0	8.7	7.2	57.2	56.8	1.6	3.9
2 ½	2.875	2.904	2.846	0.625	0.656	0.594	0.313	0.344	0.282	2.720	2.702	0.078	0.188
	73.0	73.8	72.3	15.9	16.7	15.1	8.0	8.7	7.2	69.1	68.6	2.0	4.8
	3.000	3.030	2.970	0.625	0.656	0.594	0.313	0.344	0.282	2.845	2.827	0.078	0.188
DN65	76.1	77.0	75.4	15.9	16.7	15.1	8.0	8.7	7.2	72.3	71.8	2.0	4.8
3	3.500	3.535	3.469	0.625	0.656	0.594	0.313	0.344	0.282	3.344	3.326	0.078	0.188
DN80	88.9	89.8	88.1	15.9	16.7	15.1	8.0	8.7	7.2	84.9	84.5	2.0	4.8
3 ½	4.000	4.040	3.969	0.625	0.656	0.594	0.313	0.344	0.282	3.834	3.814	0.083	0.188
DN90	101.6	102.6	100.8	15.9	16.7	15.1	8.0	8.7	7.2	97.4	96.9	2.1	4.8
	4.250	4.293	4.219	0.625	0.656	0.594	0.375	0.406	0.344	4.084	4.064	0.083	0.203
	108.0	109.0	107.2	15.9	16.7	15.1	9.5	10.3	8.7	103.7	103.2	2.1	5.2
4	4.500	4.545	4.469	0.625	0.656	0.594	0.375	0.406	0.344	4.334	4.314	0.083	0.203
DN100	114.3	115.4	113.5	15.9	16.7	15.1	9.5	10.3	8.7	110.1	109.6	2.1	5.2
4 ½	5.000	5.050	4.969	0.625	0.656	0.594	0.375	0.406	0.344	4.834	4.814	0.083	0.203
	127.0	128.3	126.2	15.9	16.7	15.1	9.5	10.3	8.7	122.8	122.3	2.1	5.2
	5.250	5.303	5.219	0.625	0.656	0.594	0.375	0.406	0.344	5.084	5.064	0.083	0.203
	133.0	134.7	132.6	15.9	16.7	15.1	9.5	10.3	8.7	129.1	128.6	2.1	5.2
	5.500	5.556	5.469	0.625	0.656	0.594	0.375	0.406	0.344	5.334	5.314	0.083	0.203
DN125	139.7	141.1	138.9	15.9	16.7	15.1	9.5	10.3	8.7	135.5	135.0	2.1	5.2
5	5.563	5.619	5.532	0.625	0.656	0.594	0.375	0.406	0.344	5.395	5.373	0.084	0.203
	141.3	142.7	140.5	15.9	16.7	15.1	9.5	10.3	8.7	137.0	136.5	2.1	5.2
	6.000	6.056	5.969	0.625	0.656	0.594	0.375	0.406	0.344	5.830	5.808	0.085	0.219
	152.4	153.8	151.6	15.9	16.7	15.1	9.5	10.3	8.7	148.1	147.5	2.2	5.6
	6.250	6.313	6.219	0.625	0.656	0.594	0.375	0.406	0.344	6.032	6.002	0.109	0.246
	159.0	160.4	158.0	15.9	16.7	15.1	9.5	10.3	8.7	153.2	152.5	2.8	6.3
	6.500	6.563	6.469	0.625	0.656	0.594	0.375	0.406	0.344	6.330	6.308	0.085	0.219
	165.1	166.7	164.3	15.9	16.7	15.1	9.5	10.3	8.7	160.8	160.2	2.2	5.6
6	6.625	6.688	6.594	0.625	0.656	0.594	0.375	0.406	0.344	6.455	6.433	0.085	0.219
DN150	168.3	169.9	167.5	15.9	16.7	15.1	9.5	10.3	8.7	164.0	163.4	2.2	5.6
	8.000	8.063	7.969	0.750	0.781	0.719	0.438	0.469	0.407	7.816	7.791	0.092	0.238
	203.2	204.8	202.4	19.1	19.8	18.3	11.1	11.9	10.3	198.5	197.9	2.4	6.1

REFERENCE NOTES ON PAGE 6

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1.1 DIMENSIONS (CONTINUED)

OGS CUT GROOVE SPECIFICATIONS FOR CARBON STEEL AND OTHER IPS PIPE



Exaggerated for Clarity

Nominal Size inches DN	inches/millimeters												
	Pipe Outside Diameter[1]			Gasket Seat "A"[2]			Groove Width "B"[3]			Groove Diameter "C"[4]		Groove Depth "D"[5] (ref.)	Min. Nom. Wall Thick. "T"[6]
	Actual	Max.	Min.	Basic	Max.	Min.	Basic	Max.	Min.	Max.	Min.		
#	8.515 216.3	8.578 217.9	8.484 215.5	0.750 19.1	0.781 19.8	0.719 18.3	0.438 11.1	0.469 11.9	0.407 10.3	8.331 211.6	8.306 211.0	0.092 2.4	0.238 6.1
8 DN200	8.625 219.1	8.688 220.7	8.594 218.3	0.750 19.1	0.781 19.8	0.719 18.3	0.438 11.1	0.469 11.9	0.407 10.3	8.441 214.4	8.416 213.8	0.092 2.4	0.238 6.1
	10.000 254	10.063 255.6	9.969 253.2	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11.9	9.812 249.2	9.785 248.5	0.094 2.4	0.250 6.4
#	10.528 267.4	10.591 269.0	10.497 266.6	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11.9	10.340 262.6	10.313 262.0	0.094 2.4	0.250 6.4
10 DN250	10.750 273	10.813 274.7	10.719 272.3	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11.9	10.562 268.3	10.535 267.6	0.094 2.4	0.250 6.4
	12.000 304.8	12.063 306.4	11.969 304.0	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11.9	11.781 299.2	11.751 298.5	0.109 2.8	0.279 7.1
#	12.539 318.5	12.602 320.1	12.508 317.7	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11..9	12.321 313.0	12.291 312.2	0.109 2.8	0.279 7.1
12 DN300	12.750 323.9	12.813 325.5	12.719 323.1	0.750 19.1	0.781 19.8	0.719 18.3	0.500 12.7	0.531 13.5	0.469 11.9	12.531 318.3	12.501 317.5	0.109 2.8	0.279 7.1
14* DN350	14.000 355.6	14.063 357.2	13.969 354.8	0.938 23.8	0.969 24.6	0.907 23.0	0.500 12.7	0.531 13.5	0.469 11.9	13.781 350.0	13.751 349.3	0.109 2.8	0.281 7.1
	14.843 377.0	14.937 379.4	14.811 376.2	0.938 23.8	0.969 24.6	0.907 23.0	0.500 12.7	0.531 13.5	0.469 11.9	14.611 371.1	14.581 370.4	0.116 2.9	0.315 8.0
15 DN380	15.000 381.0	15.063 382.6	14.969 380.2	0.938 23.8	0.969 24.6	0.907 23.0	0.500 12.7	0.531 13.5	0.469 11.9	14.781 375.4	14.751 374.7	0.109 2.8	0.312 7.9
16* DN400	16.000 406.4	16.063 408.0	15.969 405.6	0.938 23.8	0.969 24.6	0.907 23.0	0.500 12.7	0.531 13.5	0.469 11.9	15.781 400.8	15.751 400.1	0.109 2.8	0.312 7.9
	16.772 426.0	16.866 428.4	16.740 425.2	0.938 23.8	0.969 24.6	0.907 23.0	0.500 12.7	0.531 13.5	0.469 11.9	16.514 419.5	16.479 418.6	0.129 3.3	0.335 8.5
18* DN450	18.000 457.2	18.063 458.8	17.969 456.4	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	17.781 451.6	17.751 450.9	0.109 2.8	0.312 7.9
	18.898 480.0	18.992 482.4	18.863 497.1	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	18.626 473.1	18.591 472.2	0.136 3.5	0.354 9.0
20* DN500	20.000 508.0	20.063 509.6	19.969 507.2	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	19.781 502.4	19.751 501.7	0.109 2.8	0.312 7.9
	20.866 530.0	20.960 532.4	20.835 529.2	1.000 25.4	1.031 26.2	0.969 24.6	0.500 12.7	0.531 13.5	0.469 11.9	20.572 522.5	20.537 521.6	0.147 3.7	0.354 9.0
22* DN550	22.000 558.8	22.063 560.4	21.969 558.0	1.000 25.4	1.031 26.2	0.969 24.6	0.563 14.3	0.594 15.1	0.532 13.5	21.656 550.1	21.626 549.3	0.172 4.4	0.375 9.5
	22.835 580.0	22.929 582.4	22.803 579.2	1.000 25.4	1.031 26.2	0.969 24.6	0.563 14.3	0.594 15.1	0.532 13.5	22.488 571.2	22.457 570.4	0.172 4.4	0.375 9.5
24*	24.000 609.6	24.063 611.2	23.969 608.8	1.000 25.4	1.031 26.2	0.969 24.6	0.563 14.3	0.594 15.1	0.532 13.5	23.656 600.9	23.626 600.1	0.172 4.4	0.375 9.5
	24.803 630.0	24.897 632.4	24.772 629.2	1.000 25.4	1.031 26.2	0.969 24.6	0.563 14.3	0.594 15.1	0.532 13.5	24.459 621.3	24.424 620.4	0.172 4.4	0.394 10.0

REFERENCE NOTES ON PAGE 6

Applies to JIS metric pipe sizes 200A, 250A, and 300A, respectively (JIS Specification G 3452; G 3454).
* OGS cut groove specifications. For Advanced Groove System (AGS) cut groove specifications in these sizes, contact Victaulic.

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1.1 DIMENSIONS (CONTINUED)

NOTES FOR OGS CUT GROOVE SPECIFICATIONS FOR CARBON STEEL AND OTHER IPS PIPE

The cut grooving process removes less metal to less depth than threading, thus maintaining desired pipe integrity.

The coating thickness applied to the gasket sealing surface and within the groove on the cut-grooved pipe exterior shall not exceed 0.010 inch/0.25 mm. This pipe coating thickness will affect the cut groove specifications listed in the tables on pages 5 – 6. Allowances shall be made for the following:

- Pipe Outside Diameter, Gasket Seat “A”, and Groove Diameter “C” will be INCREASED by 0.020 inch/0.50 mm.
- Groove Width “B” will be REDUCED by 0.020 inch/0.50 mm.

In addition, coatings that are applied to the interior surfaces of Victaulic grooved- and plain-end pipe couplings shall not exceed 0.010 inch/0.25 mm. This includes the bolt pad mating surfaces.

For 22 – 24"/DN550 – DN600 pipe sizes, a 0.563"/14.30 mm groove width is required to obtain the maximum allowable pipe end movement listed in performance data charts for the applicable coupling. A 0.500"/12.70 mm groove width will provide half the maximum allowable pipe end movement listed in performance data charts for the applicable coupling.

1 Outside diameter: Nominal Pipe Size (ANSI B36.10) and Basic Metric Pipe Size (ISO 4200) – The average pipe outside diameter shall not vary from the specifications listed in the tables on pages 5 – 6. Maximum allowable pipe ovality shall not vary by more than 1%. Greater variations between the major and minor diameters will result in difficult coupling assembly.
Victaulic recommends square-cut pipe. Beveled-end pipe may be used, provided that the wall thickness is standard wall (ANSI B36.10) or less and that the bevel meets ANSI B16.25 (37 1/2°) or ASTM A-53 (30°). For OGS, the maximum allowable tolerance from square-cut pipe ends (measured from the true square line) is: 1/32"/0.8 mm for ¾ – 3 ½"/DN20 – DN90 sizes and 1/16"/1.6 mm for 4 – 24"/DN100 – DN600 sizes. **Square-cut pipe SHALL be used with Victaulic products containing Flush-Seal™ gaskets.**

2 "A" Dimension: The “A” dimension, or the distance from the pipe end to the groove, identifies the gasket seating area. This area between the groove and the pipe end shall be generally free from indentations, projections, weld seam anomalies, and roll marks to ensure a leak-tight seal. All oil, grease, loose paint, rust, scale, dirt, and cutting particles shall be removed.

3 "B" Dimension: The “B” dimension, or groove width, controls expansion, contraction, and angular deflection of flexible couplings by the distance it is located from the pipe and its width in relation to the coupling housings’ “key” width. The bottom of the groove shall be free from loose paint, rust, scale, dirt, and cutting particles that may interfere with proper coupling assembly.

4 "C" Dimension: The “C” dimension is the average diameter at the base of the groove. This dimension shall be within the diameter’s tolerance and concentric with the OD for proper coupling fit. The groove shall be of uniform depth for the entire pipe circumference.

5 "D" Dimension: The “D” dimension is the normal depth of the groove and is a reference for a “trial groove” only. Variations in pipe OD affect this dimension and shall be altered, if necessary, to keep the “C” dimension within tolerance. The groove diameter shall conform to the “C” dimension described above.

6 "T" Dimension: The “T” dimension is the lightest grade (minimum nominal wall thickness) of pipe that is suitable for cut grooving. Pipe that is less than the minimum nominal wall thickness for cut grooving may be suitable for roll grooving or adapted for Victaulic couplings by using *Vic-Ring* Adapters. *Vic-Ring* Adapters can be used in the following situations (contact Victaulic for details):

- When pipe is less than the minimum nominal wall thickness suitable for roll grooving
- When pipe outside diameter is too large to roll or cut groove
- When pipe is used in abrasive services

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2.0 NOTIFICATIONS

⚠ WARNING



- **Read and understand all instructions before attempting to install any Victaulic products.**
- **Always verify that the piping system has been completely depressurized and drained immediately prior to installation, removal, adjustment, or maintenance of any Victaulic products.**
- **Confirm that any equipment, branch lines, or sections of piping that may have been isolated for/during testing or due to valve closures/positioning are identified, depressurized, and drained immediately prior to installation, removal, adjustment, or maintenance of any Victaulic products.**
- **Wear safety glasses, hardhat, foot protection, and hearing protection.**

Failure to follow these instructions could result in death or serious personal injury and property damage.

3.0 REFERENCE MATERIALS

24.01: Victaulic® Pipe Preparation Tools

25.02: Victaulic® EndSeal™ Roll and Cut Groove Specifications

25.04: Victaulic® Double Cut Groove Specifications

25.09: Victaulic® Advanced Groove System (AGS™) Roll Groove Specifications

25.14: Victaulic® Innovative Groove System (IGS™) Roll and Cut Groove Specifications

25.06: Victaulic® Copper Tubing Roll Groove Specifications

29.01: Victaulic® Terms and Conditions of Sale

I-100: Victaulic® Field Installation Handbook

User Responsibility for Product Selection and Suitability

Each user bears final responsibility for determining the suitability of Victaulic products for their end-use application, in accordance with industry standards, project specifications, and Victaulic's published performance, maintenance, and safety data, as well as all warnings and installation instructions. Nothing in this or any other document, nor any verbal recommendation, advice, or opinion from any Victaulic employee, shall be deemed to alter, vary, supersede, or waive any provision of Victaulic Company's standard conditions of sale, warranty, installation instructions, or this disclaimer.

Installation

Always refer to and follow the Victaulic Installation Handbook or installation instructions for the product you are installing. Handbooks are included with each shipment of Victaulic products, providing complete installation and assembly data, and are available in PDF format on our website at victaulic.com.

Warranty

Refer to the Warranty section of the current Price List or contact Victaulic for details.

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